2) Compute the following

- Stride of 1, Filter = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Row 1 = C(1,1) = (20+35+35)+(29+44)+(16+25+21) = 225

C(1,2) = (35+35+35)+(46+42)+(25+21+19) = 258

C(1,3) = (35+35+35)+(44+42)+(21+19+19) = 260

C(1,4) = (35+35+20)+(42+27)+(19+19+12) = 209

Row 2 = C(2,1) = (29+46+44)+(16+21)+(66+120+116) = 458

C(2,2) = (46+44+42)+(25+19)+(120+116+154) = 566

C(2,3) = (44+42+42)+(21+19)+(116+154+114) = 552

C(2,4) = (42+42+27)+(19+12)+(154+114+62) = 472

Row 3 = C(3,1) = 708, C(3,2) = 981, C(3,3) = 887, C(3,4) = 802

Row 4 = C(4,1) = 1000, C(4,2) = 1488, C(4,3) = 1320, C(4,4) = 1224

$$\therefore C = \begin{bmatrix} 225 & 258 & 250 & 209 \\ 458 & 566 & 552 & 472 \\ 708 & 981 & 887 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

3) Take the output from 1. and apply a max pooling layer with, Stride = [2,2], window shape = [2,2]

Now,

$$W_{out} = \left(\frac{W_{in} - F + 2P}{2}\right) + 1$$

$$= \left(\frac{4 - 2 + 2\times 0}{2}\right) + 1 = 2$$

Output shape = 2×2

Max pooling (Block 1) $\begin{bmatrix} 225 & 258 \\ 458 & 566 \end{bmatrix} = 566$

(Block 2) $\begin{bmatrix} 250 & 209 \\ 552 & 472 \end{bmatrix} = 552$, (Block 3) = 1488, (Block 4) = 1320

∴ Final output = $\begin{bmatrix} 566 & 552 \\ 1488 & 1320 \end{bmatrix}$

4) 2×2 filter, stride 2

$$W_{out} = \left(\frac{4 - 2 + 2\times 0}{2}\right) + 1 = 2$$

∴ Output shape = 2×2

Average pooling (Block 1) $\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} = \frac{1+2+5+6}{4} = 3.5$

(Block 2) $= \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix} = 5.5$

(Block 3) $= \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix} = 11.5$

(Block 4) $= \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix} = 13.5$

∴ Final Output = $\begin{bmatrix} 3.5 & 5.5 \\ 11.5 & 13.5 \end{bmatrix}$

Exercise-2

1.a) CONV2D layer 1

K=16, F=3, Color channel=3

= (F x F x Cin + 1) x k

= 3 x 3 x 3 + 1 x 16

= 448

b) Batch Normalization layer-2

2 x 16 = 32

c) ~~CONV2D Layer 2~~ Max pooling

32F, 3x3, input = 16x16

Value = 0

d) ~~Batch normalization~~ CONV2D layer-2

= (3x3 x16 + 1) x 32

= 4610

e) Batch normalization layer 2

2 x No. of channel = 2 x 32 = 64

f) Max pooling layer-2

0

g) Flattening

- 8x8x32 = 2048

h) Dense layer 1

Weight = input x neurons = 2048 x 128 = 262,144

i) W = 128x10 = 1280

b = 10

Total = 1280 + 10 = 1290